Game Design Document

Chainy: A 3D World Educational Game for learning emerging Technology (Blockchain & AI)



Offline foundational learning for ages 12-17

Web3, blockchain and AI through short guided activities, with particular attention to learners in low-income and remote communities.

Illustrated design proposal | 8 October 2026

Concept art and interface diagrams explain the intended experience. They are not screenshots of an implemented game.

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About this document

This document explains what the learner sees, does and learns. Numbered sections describe the game brief, gameplay, worlds, presentation, mechanics, characters and delivery plan. Illustrations sit with the decisions they explain.

Detailed implementation rules are kept in the companion mechanics reference. The confirmed project mission and scope remain in the project baseline. Proposed lesson quantities, names and timing must be checked during development.

1. Game objectives and brief

1.1 Game brief

Chainy introduces essential ideas about **Web3, Blockchain and AI** through short illustrated lessons and puzzles. It is designed especially for learners in low-income and remote communities with limited access to technology. The game assumes neither previous knowledge of these subjects nor personal ownership of a phone.

The player taps one of three worlds, meets a friendly guide and works through an everyday situation. They might decide whether a message is safe, put delivery records in order, compare different copies of a record, or help a robot deliver a parcel. Each choice produces an explanation and a chance to try again.

Genre: Single-player educational lesson-and-puzzle game. **Player:** One learner at a time, with separate local profiles for shared devices. **Game elements:** Choosing, sorting, ordering, linking, comparing, predicting, revising and answering short questions.

1.2 Narrative brief

A blue-haired explorer in a yellow coat welcomes the learner to a small technology learning community. Its three worlds make unfamiliar ideas visible: Web3 World explores information and safe choices; Blockchain World examines records and shared copies; AI World investigates examples, predictions and actions.

Practical stories motivate each lesson: a club reviews who can change its list, a delivery record has been altered, or a sorting helper makes a mistake. The learner inspects the situation and decides what to do.

1.3 Game objectives

The intended learner can:

- Ask who controls information and recognize requests that should be checked carefully.
- Explain why records need an order and how a saved reference helps reveal a change.
- Compare three copies using the lesson's stated rule and recognize when that rule cannot choose an answer.
- Distinguish examples used for learning from new examples used for checking a prediction.
- Explain a simple robot's observation, chosen action and result.

Key Features: Offline access, three themed worlds, short guided challenges, helpful retry feedback and saved progress.

Finishing a lesson is evidence of completing its activities, not proof of expertise. Actual learner understanding, readability and enjoyment must be evaluated with the intended audience.

2. Gameplay

2.1 Playing Chainy

You open Chainy without connecting to the internet. After choosing your learner profile, the explorer welcomes you to a colorful map. You tap Blockchain World and open a lesson about delivery records. The guide shows a short example, then gives you three new cards: Pack, Dispatch and Receive.

You place Receive before Dispatch and press Check. The game highlights the problem: "This receipt has no earlier dispatch." You inspect the cards, change the order and connect the records. The guide explains why your correction works. Three short questions ask you to apply the idea again. Once your result is saved, the lesson card shows your progress and you can return to the map.

2.2 Opening, options and modes

The opening menu offers **Play**, **Continue** and **Settings**. On a shared phone, the teacher provides a learner code so each person can select their own local progress. Entering the same code on another phone does not bring an earlier save to that phone.

Play opens the world map. Continue returns to the learner's last saved point. Settings include sound and readability options. Pause offers Resume, Return to Map and Switch Learner. Switching clearly identifies that the current session is ending.

There is one guided learning journey, with replay of completed lessons. Teacher export is a separate staff activity. No multiplayer, online account, purchase or cryptocurrency transaction is needed.

2.3 The lesson loop



Figure 1. Proposed lesson flow. A mistaken answer returns the learner to an explanation and another attempt.

Navigation is described in section 2.2 and the worlds in section 3; activities appear in section 5, with completion checks and rewards in sections 2.4 and 5.5.

A typical lesson follows six steps: choose a lesson; read a short explanation; inspect an example; try the activity; review the outcome; answer fresh questions and save. Help remains available. Learners can pause rather than rush through an unfamiliar term.

2.4 Winning, losing and the ending

A lesson is completed after its required activity and understanding check. The proposed check contains three questions: answer all three and correctly answer at least two, with explanations and retries. Repeating one correct question does not count as answering another. The result must be saved before completion appears.

There is no death, health loss or permanent failure. A wrong choice, broken link or blocked robot is an invitation to inspect and revise. Previously earned progress is retained. Completing the agreed foundation lessons opens a journey summary, followed by replay or exit.

2.5 Why the game should be enjoyable

Short challenges, predictions and visible consequences encourage curiosity. Revision gives the learner control over improvement. Friendly characters and progress support the journey. These intentions need learner testing.

3. Game world

3.1 The three-world map



Figure 2. Proposed map presentation. Tap a world to open its lesson cards; the map does not require walking or a joystick.

All three worlds are visible. Their first lessons are available after a brief touch-control introduction. Later lessons unlock in order within each world; completed lessons can be replayed. Web3 is a useful starting point, but learners do not have to finish it before opening Blockchain or AI.

The **nine-lesson draft** is a proposal, not a confirmed count. The proposed three-to-five-minute lessons need learner testing, including reading support and retries.

3.2 Web3 World

Familiar community examples introduce control, privacy and safe decisions. Web3 scenery follows Chainy's rounded style; its details remain proposed.

Proposed lesson	What the learner does
Who Controls the Information?	Compare a list controlled by one person with shared responsibility, then ask who can approve changes.
Public Information and Secrets	Sort fictional information into appropriate sharing choices; distinguish a receiving address from a secret.
Pause and Check	Inspect suspicious messages, recognize pressure or secret requests, and choose a safe next step.

The game uses pretend information. It never asks learners to enter real passwords, private keys, home addresses or payment details.

3.3 Blockchain World

Blue and green island scenery frames clear record cards and friendly robot guidance. The learning area focuses attention on events, links and copies rather than a large landscape.

Proposed lesson	What the learner does
Put the Records in Order	Arrange Pack, Dispatch and Receive using a stated rule, then link neighboring records.
Find the Changed Record	Compare a current record with a saved reference, identify the difference and check a correction.
Compare Three Copies	Inspect three copies, preview a repair and recognize a case with no agreement.

The examples explain limited classroom rules. They do not claim that matching copies or a hash proves an event happened in the real world.

3.4 AI World

Violet scenery and cyan highlights distinguish AI World while preserving the same characters and interface. Fictional parcels make examples, features and decisions concrete.

Proposed lesson	What the learner does
Teach with Examples	Label practice parcels, train a small sorting model and test a fresh parcel.
Learn from a Mistake	Inspect an incorrect label, revise the examples and compare the changed prediction.
Guide a Simple Agent	Arrange a short sequence of commands and help a robot deliver a parcel on a small grid.

The sorting activity must genuinely learn from the supplied examples and work clearly on a phone before it becomes production content. It must not present fixed answers as learned predictions.

4. Art, interface, storyboard and sound

4.1 Art direction

The first two supplied Chainy storyboards are the visual authority: rounded toy-like 3D characters, soft lighting, a blue-haired explorer in yellow, friendly screen-faced robots, chunky scenery and welcoming colors. Keep this identity while using respectful explanations suited to ages 12-17.

The third, BlockForge image contributes gameplay ideas only. Its flat illustration style, title, money rewards and certificate are not adopted. Likewise, the fantasy setting, combat and keyboard controls of *Project Vanishing* are not Chainy features; that document guides this outline and illustrated presentation.

Artwork highlights the current action. Final instructions, quantities and answers come from reviewed lesson content, not decorative text.

4.2 Interface and controls

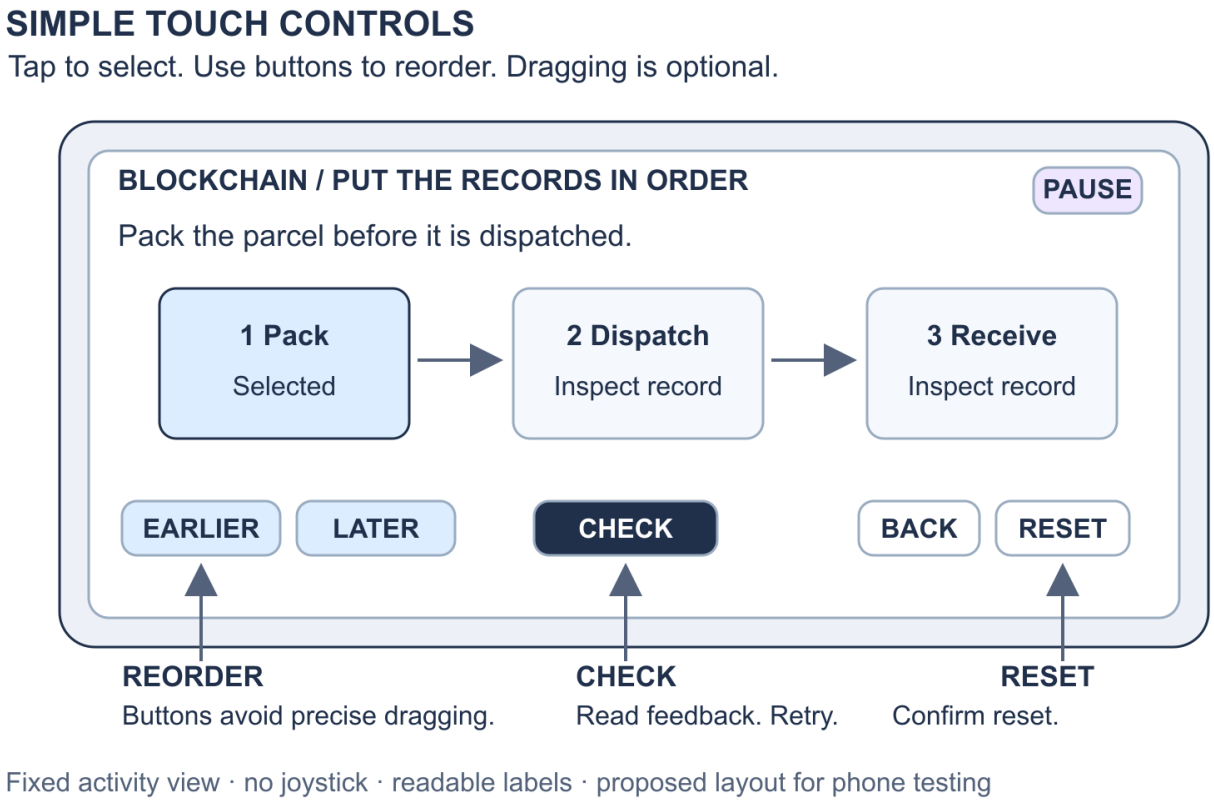


Figure 3. Proposed touch interface. Clearly labeled buttons provide an alternative to dragging.

The learner taps world and lesson cards, advances short explanations and selects an answer. Ordering uses selectable cards with earlier/later controls; dragging may be offered as an additional choice. A clear Check button tests the current answer. Back, Help and Pause remain easy to find.

The main screens are the learner profile, menu, world map, lesson introduction, activity, feedback, understanding check, progress summary and settings. Staff export is visually separate. Important text must fit on the phone with larger text enabled; colors always have labels or symbols to explain their meaning.

4.3 Storyboard

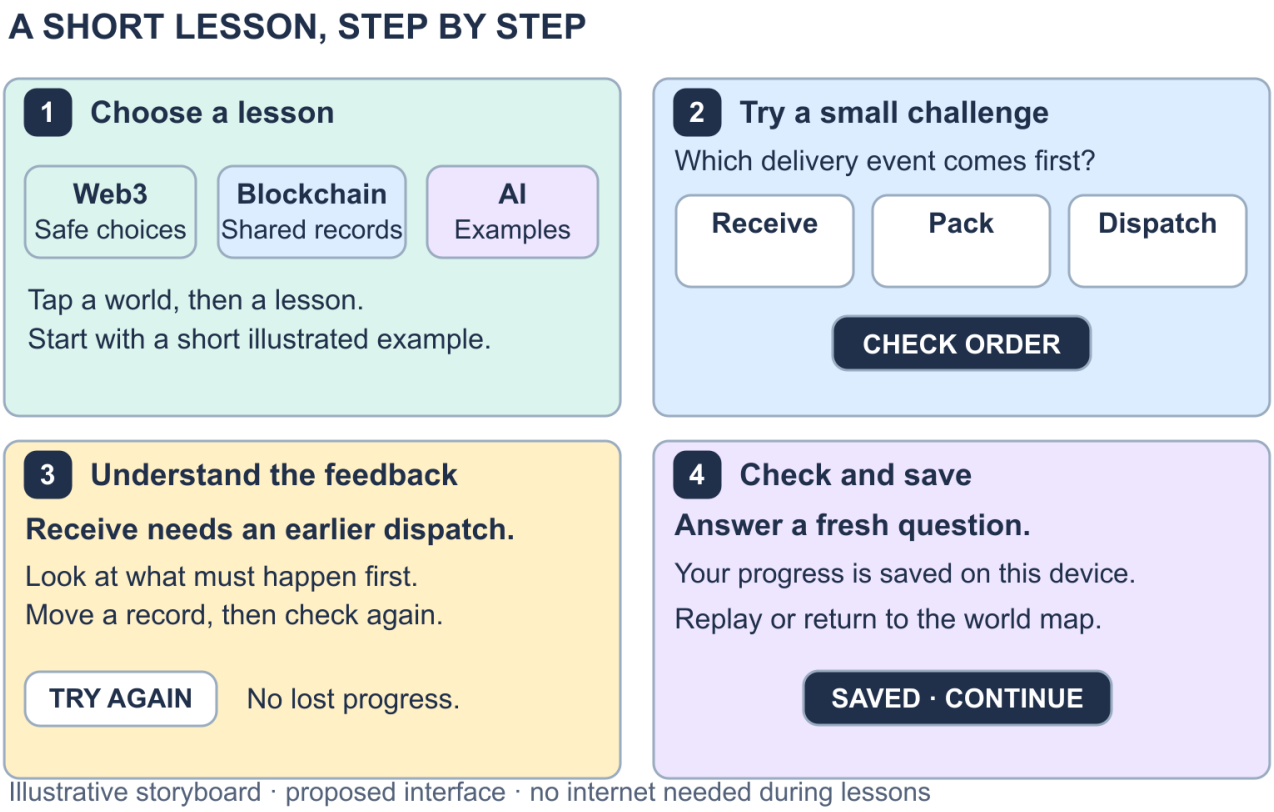


Figure 4. Proposed player journey. Each scene serves one step of the learning experience.

The panels show welcome, objective/example, learner activity, explanatory feedback, a fresh check and saved progress. Reusing this sequence lets learners focus on each new concept without relearning the interface.

4.4 Sound and accessibility

Use short sounds for selecting an item, checking an answer and completing a lesson. Optional gentle music may establish the world's mood. Sounds remain secondary to visible feedback, and every required instruction works with audio muted. Narration is not assumed.

Begin with an unscored touch practice. Use plain English, one instruction at a time and meaningful pictures. A teacher may read instructions without choosing answers for the learner. Do not confuse unfamiliarity with a phone, slow reading or asking for help with poor understanding of the concept.

5. Game mechanics

5.1 Lesson guidance and choices

A guide presents a goal, one worked example and a small task. The learner predicts an outcome before checking it. Scenario activities ask the learner to compare, select or sort fictional items. Feedback names the reason: "This message asks for a secret," or "This receipt needs an earlier dispatch." It does not merely show a red cross.

5.2 Records, changes and copies

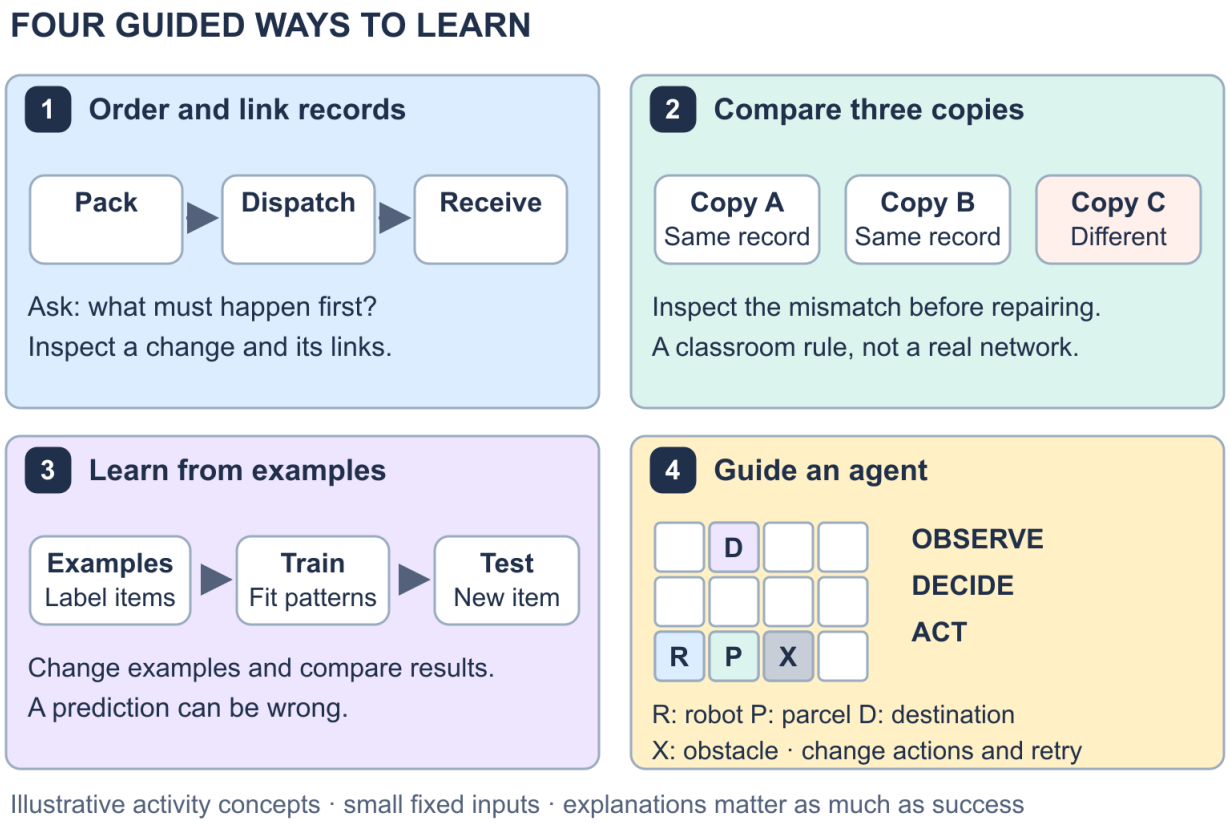


Figure 5. The core activities turn abstract ideas into visible choices and consequences.

For record ordering, the learner uses the supplied cards and links. Pack must come before Dispatch, and Dispatch before Receive. This is a bounded puzzle; free placement of an unlimited set of components is deferred.

For change detection, a saved reference is shown beside the current record. A changed destination or quantity creates a visible mismatch. Restoring it repairs the exercise, but the explanation distinguishes a matching record from proof that a real delivery occurred.

For comparing copies, three panels represent copies stored on the same device. In the normal exercise, two matching valid copies suggest how to repair one changed copy. A separate case with three different copies shows why the simple rule cannot decide and needs the provided reference. No internet connection or real distributed network is involved.

5.3 Learning from examples

The learner labels fictional parcels using visible characteristics, such as weight and fragility. Train uses those examples to form a small model. Test then asks about a parcel kept separate from the practice examples.

A deliberately incorrect label allows the learner to see a wrong or uncertain prediction. They inspect the examples, correct the label, train again and compare. More examples do not automatically mean better results. The lesson keeps the model's prediction separate from the guide's comparison answer and explains when the result is uncertain.

5.4 The delivery agent

A small fixed grid contains a robot, parcel, obstacle and destination. Available commands include Move, Turn, Pick Up, Deliver and Stop. The learner arranges a short sequence, predicts the next action and presses Step or Run.

A move into an obstacle stops with an explanation. Reaching the destination alone is insufficient: the robot must carry and deliver the parcel. The learner can revise and restart. This robot follows visible commands; it does not learn its route from the parcel-sorting lesson.

5.5 Progress, rewards and recovery

Lesson cards show available, in-progress and completed states. Badges acknowledge saved completion. They are not certificates, a currency or a public ranking. Replaying a lesson retains its earlier results and earned progress.

Pause and Return to Map use the latest saved point. If saving fails, the game explains the problem and offers another attempt instead of displaying false completion. Reset affects the current activity, not the learner's whole history. Teachers can export the approved learner records separately; saving an export file does not automatically transfer it or delete local progress.

6. Characters and player definition



Figure 6. Proposed interpretation of approved Chainy visual identity. Proportions and costume details require comparison with original references.

6.1 Explorer and guides

The explorer is a welcoming companion rather than a combat hero. Blue hair, a yellow coat and a small backpack provide a recognizable silhouette across the map and lesson panels. The learner controls choices and activities directly; walking this character through a world is not required.

Blocky is the proposed blockchain guide, helping the learner inspect records and copies. Aira is the proposed AI guide, encouraging predictions and careful checking. Friendly robot designs may share the same basic model. A guide can also present Web3 scenarios without requiring another full character.

6.2 Player properties and feedback

Property	How it changes and what the learner sees
Selected lesson	Choosing a card updates the objective and available activity.
Current answer or arrangement	Selecting, ordering or linking updates the visible choice before Check.
Activity outcome	Checking reveals a success explanation or a useful correction.
Saved lesson progress	Completed work changes the lesson card and may award a badge.
Sound and readability preferences	Settings give an immediate, visible or audible response.

Health and weapons do not apply. There is no combat, death penalty or equipment system. Information cards support a lesson rather than acting as health pickups. Rewards are understanding, saved progress and brief recognition of completion.

6.3 Character behavior

Use restrained gestures such as greeting, pointing, thinking and celebrating. The guide should point toward the relevant record or choice and leave room for the explanation. A short reaction may acknowledge success, but it must not interrupt correction or cover the next action.

7. Project plan

7.1 Platform and technical specifications

Field	Retained design target
Technical form and view	Rounded 3D visual identity, tappable map, fixed activity views and 2D interface; landscape is proposed.
Platform and device	Android touchscreen phone; one player at a time.
Language	Unity/C# for development; English for the first version.
Phone specification	Android 10+/11+, 4 GB physical RAM, 64 GB total storage, ARM64, OpenGL ES 3.2 or Vulkan 1.1, entry-level-to-midrange GPU.
Saves and connectivity	Local SQLite saves; required lessons work offline. Installation, teacher file transfer and updates are separate.

Android 11+ is the provisional test starting point, with Android 10 a compatibility candidate. These retained specifications are not verified support claims. Total phone storage is not the game's required space. Test actual phones before claiming support.

7.2 Team and parallel work

Sherwin Limosnero coordinates scope and delivery. Christian Jude Villaber leads technical integration. Sean Matthew Narvaez leads learning documentation and research materials. Luis Cabrido III leads artwork. Miguel De Leon supports interface work, documentation and device checks.

Parallel assistance is divided into content writing, illustrations, gameplay activities, interface work and independent review. Each worker receives a defined output and owner. The reviewer checks that pictures, instructions and actual behavior agree before integration. Shared screens and final builds have one integration owner. Balanced work must reflect available hours; the earlier 520-hour estimate has not been recalculated for the revised scope.

7.3 Milestones and evidence

Conditional checkpoint	Expected result
October 9-11	First phone build with working save, reopen and export checks.
October 18	One complete lesson and a checked example-learning activity; research preparation advanced.
October 25-30	Remaining agreed lessons brought together; integrated beta candidate targeted for October 30.
October 31-November 2	Readiness checks and correction of blocking problems.
November 3-6	Possible beta at Tambunan National High School, Marinduque, subject to booking and required arrangements.
November 7-16	Fixes, evidence review, analysis and manuscript completion.
November 17-20	Rehearsal, backups and critical corrections.
November 21, 2026	**Confirmed pre-final defense.**

The beta is an opportunity, not a confirmed appointment or completed evaluation. Lesson count and schedule depend on actual production and phone checks. A successful document or illustration does not prove a working game. The supplied rubric and its 32-point structure remain; the school's method for measuring 60-percent completion still needs clarification. Lessons or task counts cannot establish that percentage.

8. References and supporting documents

1. **Document presentation:** The user-supplied 27-page *Project Vanishing* GDD informs the numbered outline and use of supporting visuals, not Chainy's gameplay features.
2. **Visual direction:** The first two supplied Chainy storyboards define the character and world style. The third BlockForge image supplies gameplay context only. New illustrations are design proposals.
3. **Learning content:** Selected concepts from the supplied *Web3 Encyclopedia*, checked against primary educational references; *Giggle Academy* provides a learning-format reference. Full A-Z coverage is outside the agreed scope.
4. **Concept accuracy:** NIST blockchain material, Google machine-learning educational material, and relevant privacy and responsible-AI references inform the beginner explanations. Detailed source mapping is retained in the learning materials.
5. **Project decisions:** The user's approved mission, retained phone specifications, school deliverables and November 21 deadline govern this design.

For the complete lesson copy and questions, see beginner learning materials. The project baseline records confirmed decisions and the active plan owns delivery dates. Detailed rules, examples, traceability and engineering links remain in the game mechanics reference. Media provenance identifies reference and generated assets.